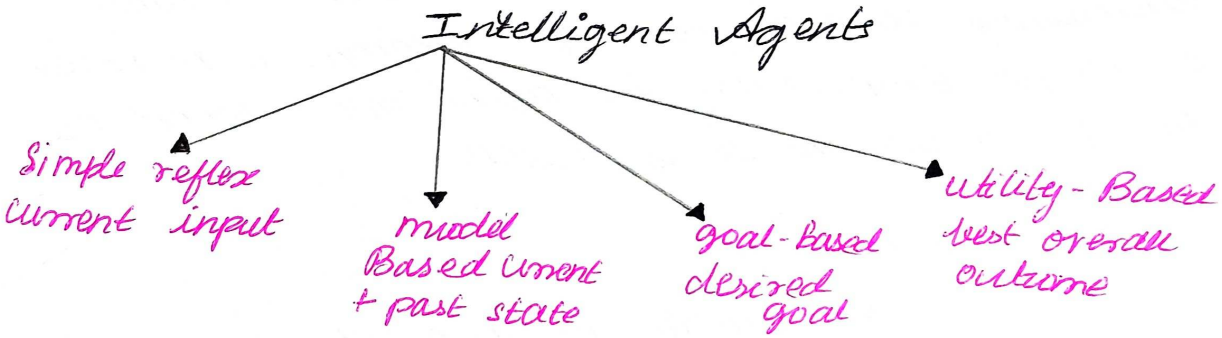
1. Smart Home Automation System and intelligent Agents

Introduction

A smart home automation system uses Artificial intelligence (AI) based system that observes its environment through sensors and controls device through actuators. inputs may include motion, temperature, time of day, door / window status and security events an intelligent agent previes these inputs makes a decision and performs an action.



∴ Main types of intelligent agent

Simple Reflex Agent :

A simple reflex agent acts only on the current percept using condition action rule.
ex. if motion is detected in a dark room. switch the light on. if temperature becomes high switch on the fan or AC. if an intrusion is detected activate the alarm.

Model - Based Reflex Agent

A model Based agent maintains an internal state representing what is happening in the home it can remember room ouppancy previous temperature readings door / window states and appliance status for example - if the AC is running while a window is open the system can switch off the AC to avoid energy waste.

Goal - Based Agent :-

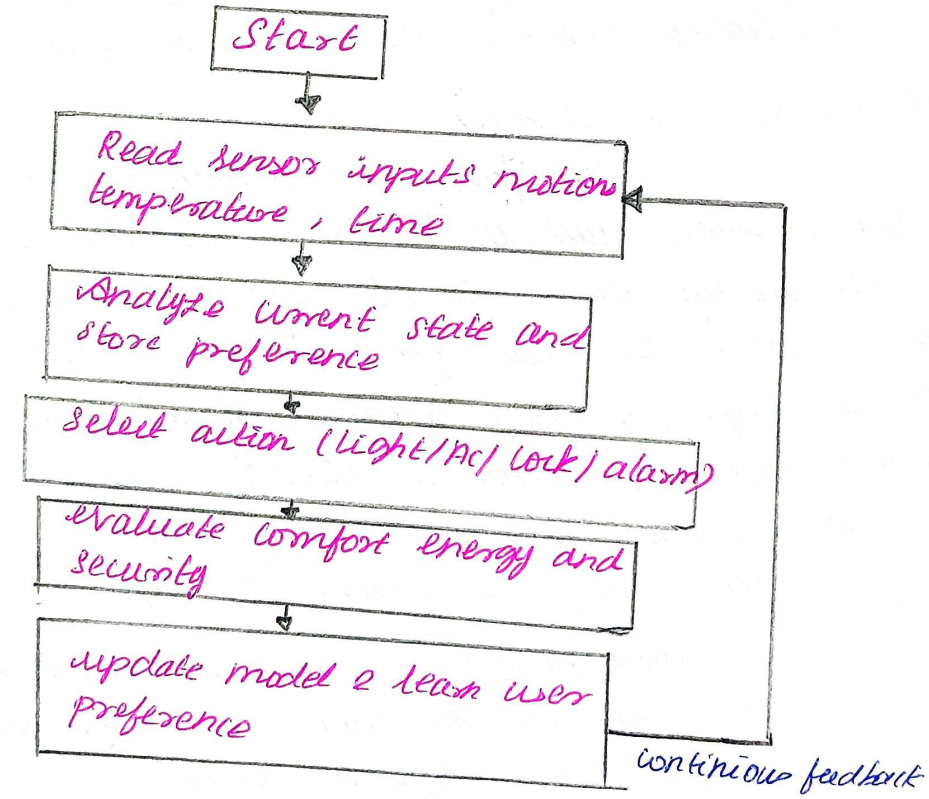
A goal-based agent chooses actions that help achieve a specified goal. smart-home goal may include maintaing a comfortable temperature keeping the home secure and reducing unnecessary electricity use. for example to maintain a target temperature the agent may control the AC fan or curtains it is flexible

Utility - Based agent

A utility - based agent assigns values to possible outcomes and selects the action that gives the greatest overall benefit in a smart home it can balance comfort energy efficiency and security for example extreme cooling may increase comfort but consume more electricity while moderate cooling can provide better overall utility its main limitation is that designing a suitable utility function can be complex.

Recommended approach :-

The most effective solution is a hybrid agent simple reflex rules can handle urgent events a model based component can remember the state of the home. a goal based component can work toward comfort and security goals and a utility-based component can balance comfort energy use and safety. A learning component can gradually adapt to the user's preferences this hybrid design provides fast response.



∴ Smart-home decision making flowchart

Conclusion :-

A hybrid intelligent agent is most suitable because no single agent type handles every requirement equally well. Combining reflex response internal state, goal, utility and learning gives a smart home better comfort, energy efficiency and security

2. Robot in an Unknown Maze

A robot is placed in an unknown maze and must reach the exit without prior knowledge of the layout uniformed search can systematically explore the available path

Uniform cost search (UCS)

uniform cost search expands the frontier node with the lowest accumulated path cost it normally uses a priority que if movement cost differ. UCS can find the least cost route. for example a short route through difficult terrain may cost more than a slightly longer route through normal terrain so UCS compares total cost rather than only the number of steps

Advantages : Complete when step costs are positive ; optimal for least-cost paths suitable when actions have different costs.

disadvantages : may expand many nodes; can require large memory : may be slow in a large maze.

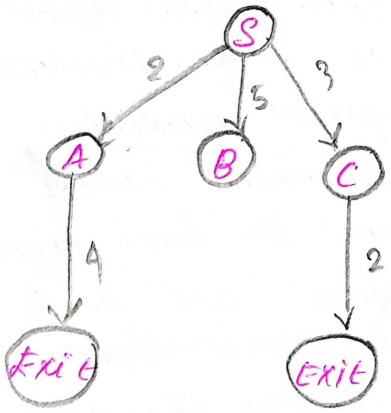
Iterative Deepening search (IDS)

Iterative deepening search repeatedly performs dept-limited DFS. it first searches dept 0 then dept 1, dept 2 and so until the exit is found it combines the low memory use of DFS with the completeness of breadth wise exploration

Advantages: complete for finite branching factor, uses little memory; optimal when every steps has equal cost

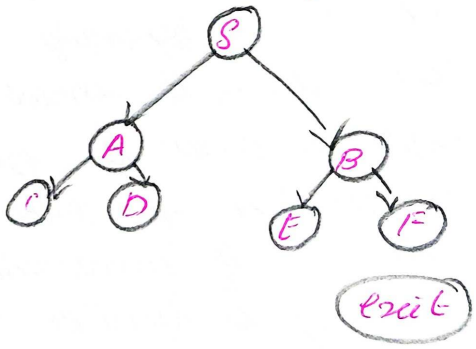
disadvantages: upper level nodes are explored repeadelly it not cost optimal when action have different costs

UCS example



least cost : 5 -> C -> exit = 5

IDS example



dept limits

0 -> 1 -> 2 -> 3 (exit found)

UCS least count and IDS dept-by-dept search

Relation to Intelligent Agents

Simple reflex: reacts to the immediate situation such as turning when a wall is detected but may repeat paths

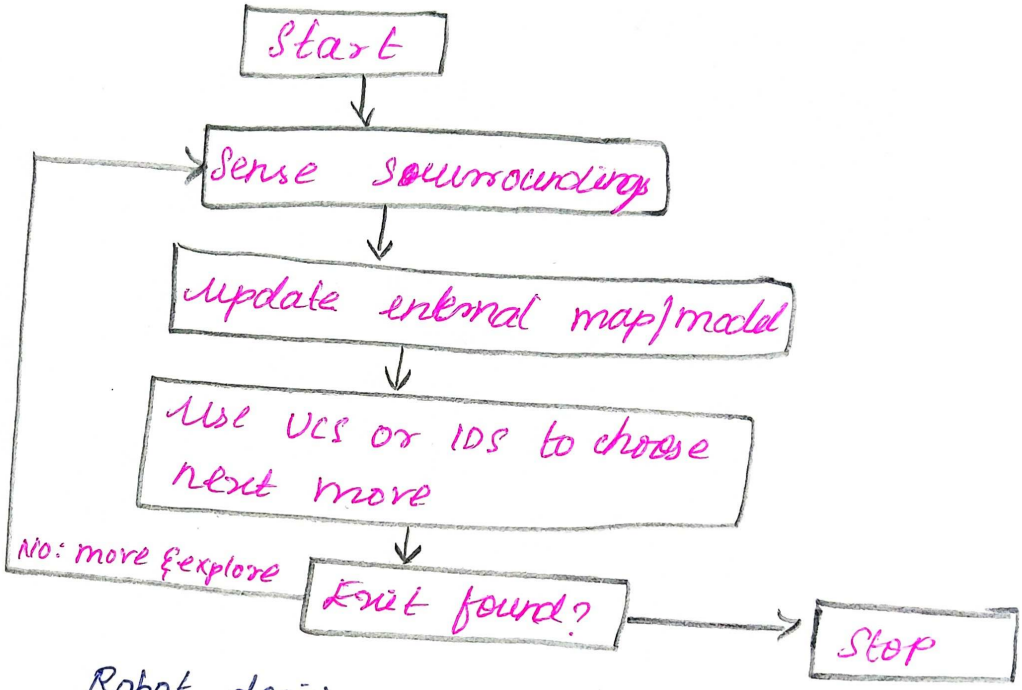
model-based: stores explored paths walls and dead ends. which is highly useful in an unknown maze

goal-based: uses the objective 'reach the exit' to guide search decisions

Utility-based: can additionally consider travel time, energy, distance and safety.

Most effective combination:

A model-based + goal-based agent with UCS is generally the best choice when movement costs differ. the model based component remembers explored areas the goal based component focuses on reaching the exit and UCS find the least count rates if all movements have equal cost and memory is limited model based + goal-based with IDS is a strong alternative because IDS uses much less space.



Robot decision-making in an unknown maze.

Conclusion :-

Agent design and search strategy should work together. In an unknown maze memory of explored areas prevents repeated work, the goal directs the robot toward the exit and UCS or IDS provides a systematic search method. UCS is preferred for varying costs. While IDS is attractive for equal-cost movement and tight memory limit